# Synthesis of Methyl Enanthate via Fischer Esterification

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03/06/26

## 1 Introduction

The objective of this experiment is to synthesize methyl enanthate via Fischer Esterification by reacting methanol with enanthic acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz $^1$H NMR, and GC-MS.

## 2 Reaction Scheme

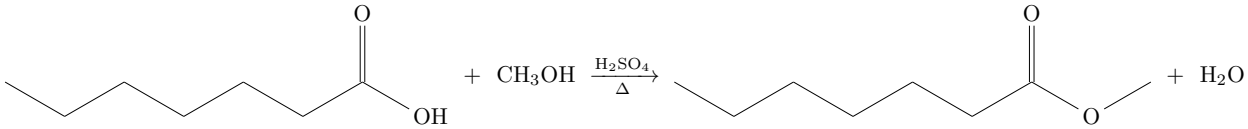



Figure 1: Reaction of Enanthic Acid and Methanol

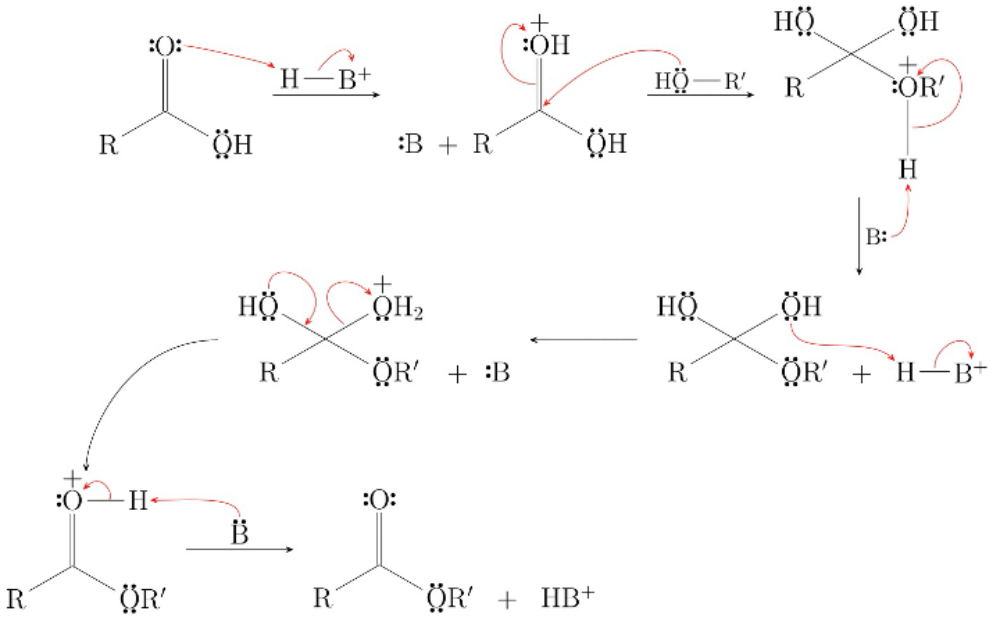



Figure 2: Fischer Esterification Mechanism

# 3 Table of Reagents and Products

| Compound | MW (g/mol) | Density (g/mL) | B.P (°C) | Equiv | Amount | mmol |
|---|---|---|---|---|---|---|
| **Enanthic Acid** | 130.18 | 0.918 | 223 | 1.0 | 7.899 g / 8.60 mL | 60.7 |
| **Methanol** | 32.04 | 0.792 | 64.7 | 3.0 | 5.832 g / 7.36 mL | 182.0 |
| **Sulfuric Acid** | 98.08 | 1.84 | – | cat. | 5 drops | – |
| **Diethyl Ether** | 74.12 | 0.710 | 34.6 | solv. | – | – |
| **Methyl Enanthate** | 144.21 | 0.875 | 171 - 173 | – | 8.75 g / 10.00 mL | 60.7 |

# 4 Experimental Procedure

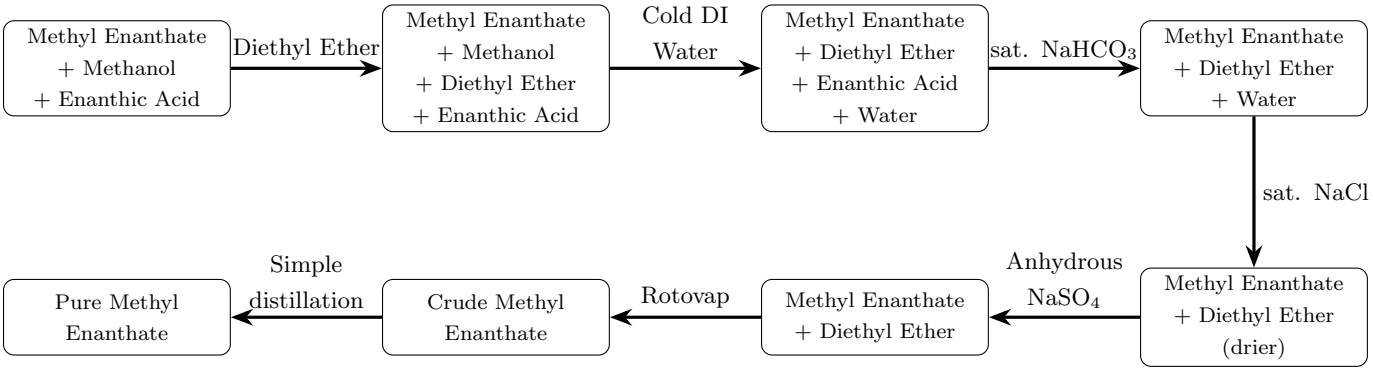
## 4.1 Apparatus

- Reflux
- Simple Distillation
- Rotovap

## 4.2 Procedure

- In a 25 mL round-bottom flask, add 7.899 g (8.60 mL) of enanthic acid followed by 5.832 g (7.36 mL) of methanol and a stir bar.
- While stirring, slowly add 5 drops of concentrated sulfuric acid.
- Gently reflux for 60 minutes, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel using 20 mL total of diethyl ether to aid in the process.
- Quench the organic layer with 15 mL of iced cold DI water and collect the organic layer. Do this three times.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of $CO_2$ gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Gently boil off the ether solution in a 50 mL beaker blowing a light stream of air over the top to obtain the crude methyl enanthate.
- Purify the product with simple distillation.
- Analyze the purified product by odor assessment, ATR-IR, $^1$H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

## 5 Purification Scheme




## 6 Results and Data

| g of Enanthic Acid | g of Methanol | Drops of $H_2SO_4$ | Reflux Time |
|---|---|---|---|
| | | | |
| Mass of Crude Product | | Mass of Purified Product | |
| Theoretical Yield | | | |

| Crude Percent Yield | Pure Percent Yield |
|---|---|
| | |
| Crude Product Observations | Purified Product Observations |
| | |